DS 592 Introduction to Sequential Decision Making	**Homework #2**
Spring 2025	Due: February 19, at 5 pm

Your name: Achuthan Rathinam
Your collaborators: None

Homework is due via Gradescope. You are encouraged to collaborate with up to three other students, but **you must write up your solutions individually**; what is submitted should represent your own words and thoughts. List your collaborators at the top (even if "none").

Problem 1

Suppose that X is σ-subgaussian and X_1 and X_2 are independent and σ_1- and σ_2-subgaussian, respectively. Then:

a. cX is $|c|\sigma$-subgaussian for all $c \in \mathbb{R}$.

(a) A r.v is σ- Gaussian $\forall$ all $\lambda \in \mathbb{R}$,

$$\mathbb{E}\left[e^{(\lambda X)}\right] = \exp\left(\frac{\lambda^2\sigma^2}{2}\right)$$

then, by scaling it c $\forall$ $c \in \mathbb{R}$

$$\mathbb{E}\left[e^{(\lambda c X)}\right] = \mathbb{E}\left[e^{(\lambda c)X}\right] \leq \exp\left(\frac{(\lambda c)^2\sigma^2}{2}\right)$$

$$\leq \exp\left(\frac{\lambda^2(|c|\sigma)^2}{2}\right) \quad \left[c^2 = |c|^2\right]$$

$\therefore$ So cX is $|c|\sigma$- subgaussian $\forall$ $c \in \mathbb{R}$.

b. $X_1 + X_2$ is $\sqrt{\sigma_1^2 + \sigma_2^2}$-subgaussian.

(b) Objective:

If X_1 - σ_1-Subgaussian

X_2 - σ_2-Subgaussian

then $X_1 + X_2$ is $(\sigma_1 + \sigma_2)$ subgaussian. (given $X_1 \perp X_2$)

For any $\lambda \in \mathbb{R}$

$$\mathbb{E}\left[e^{\lambda(X_1+X_2)}\right] = \mathbb{E}\left[e^{\lambda X_1} \cdot e^{\lambda X_2}\right]$$

By Independence,

$$\mathbb{E}\left[e^{\lambda X_1} \cdot e^{\lambda X_2}\right] = \mathbb{E}\left[e^{\lambda X_1}\right] \mathbb{E}\left[e^{\lambda X_2}\right]$$

Applying the subgaussian bounds

$$\mathbb{E}\left[e^{\lambda X_1}\right] \mathbb{E}\left[e^{\lambda X_2}\right] \leq \exp\left(\frac{\lambda^2 \sigma_1^2}{2}\right) \exp\left(\frac{\lambda^2 \sigma_2^2}{2}\right)$$

$$\leq \exp\left(\frac{\lambda^2 (\sigma_1^2 + \sigma_2^2)}{2}\right)$$

Comparing it to $\exp\left(\frac{\lambda^2 \sigma_{new}^2}{2}\right) \Rightarrow \sigma_{new}^2 = \sigma_1^2 + \sigma_2^2$

$\therefore \sigma = \sqrt{\sigma_1^2 + \sigma_2^2}$

$\therefore$ $X_1 + X_2$ is $\sqrt{\sigma_1^2 + \sigma_2^2}$ subgaussian

Problem 2

Let X_i be σ_i-subgaussian for $i \in \{1, 2\}$ with $\sigma_i \geq 0$. Prove that $X_1 + X_2$ is $(\sigma_1 + \sigma_2)$-subgaussian. Do not assume independence of X_1 and X_2.

Hint: Consider the Hölder inequality; for real-valued random variables X, Y it is true that $\mathbb{E}[|XY|] \leq (\mathbb{E}[|X|^p])^{1/p} (\mathbb{E}[|Y|^q])^{1/q}$, where $\frac{1}{p} + \frac{1}{q} = 1$ and $p, q > 1$.

2) X_i is σ_i-subgaussian

ie $\forall \lambda \in \mathbb{R}$ $\mathbb{E}[e^{\lambda X_i}] \leq \exp\left(\frac{\lambda^2 \sigma_i^2}{2}\right)$

Fix $\lambda \in \mathbb{R}$.

Let

$U = e^{\lambda X_1}$, $V = e^{\lambda X_2}$

Then,

$UV = e^{\lambda(X_1+X_2)}$ $\quad U, V \geq 0$.

By Holder Inequality

$p, q > 1$ & $1/p + 1/q = 1$

$$\mathbb{E}[e^{\lambda(X_1+X_2)}] = \mathbb{E}[UV] \leq (\mathbb{E}[U^p])^{1/p} (\mathbb{E}[V^q])^{1/q} = (\mathbb{E}[e^{p\lambda X_1}]^{1/p})(\mathbb{E}[e^{q\lambda X_2}])^{1/q}$$

Applying subgaussianity of X_1 & X_2

$$(\mathbb{E}[e^{p\lambda X_1}])^{1/p} \leq \exp\left(\frac{(p\lambda)^2 \sigma_1^2}{2p}\right) = \exp\left(\frac{\lambda^2 p \sigma_1^2}{2}\right)$$

$$\text{|||}^{ly} \quad (\mathbb{E}[e^{q\lambda X_2}])^{1/q} \leq \exp\left(\frac{\lambda^2 q \sigma_2^2}{2}\right)$$

Hence,

$$\mathbb{E}[e^{\lambda(X_1+X_2)}] \leq \exp\left(\frac{\lambda^2}{2}(p\sigma_1^2 + q\sigma_2^2)\right)$$

Choosing

$$p = \frac{\sigma_1+\sigma_2}{\sigma_1} \quad \& \quad q = \frac{\sigma_1+\sigma_2}{\sigma_2} \quad \left[1/p + 1/q = 1\right]$$

Substituting we get,

$$\mathbb{E}[e^{\lambda(X_1+X_2)}] \leq \exp\left(\frac{\lambda^2}{2}(p\sigma_1^2 + q\sigma_2^2)\right)$$

$$\leq \exp\left(\frac{\lambda^2}{2}\left(\left(\frac{\sigma_1+\sigma_2}{\cancel{\sigma_1}}\right)\sigma_1^{\cancel{2}} + \left(\frac{\sigma_1+\sigma_2}{\cancel{\sigma_2}}\right)\sigma_2^{\cancel{2}}\right)\right)$$

$$\leq \exp\left(\frac{\lambda^2(\sigma_1+\sigma_2)^2}{2}\right)$$

$\therefore$ $X_1 + X_2$ is $(\sigma_1 + \sigma_2)$ - Subgaussian

Problem 3

Let $X_1, \ldots, X_n$ be a sequence of σ-subgaussian random variables (possibly dependent) and $Z = \max_{t \in [n]} X_t$. Prove that

a. $\mathbb{E}[Z] \leq \sqrt{2\sigma^2 \log(n)}$.

3) Given:-

Let $X_1, \ldots, X_n$ be a σ-subgaussian

Let

$$Z = \max_{t \in [n]} X_t$$

3a) Bound on $\mathbb{E}[Z]$

For any $\lambda > 0$,

$$e^{\lambda Z} = e^{\lambda \max_t X_t} = \max_t e^{\lambda X_t} \leq \sum_{t=1}^{n} e^{\lambda X_t}$$

then,

$$\mathbb{E}[e^{\lambda Z}] \leq \sum_{t=1}^{n} \mathbb{E}\left[e^{\lambda X_t}\right] \leq n \exp\left(\frac{\lambda^2 \sigma^2}{2}\right)$$

By jensen,

$$e^{\lambda \mathbb{E}[Z]} = \mathbb{E}\left[e^{\lambda Z}\right]$$

Hence,

$$e^{\lambda \mathbb{E}[Z]} = n \exp\left(\frac{\lambda^2 \sigma^2}{2}\right)$$

$$\lambda \mathbb{E}[Z] = \log n + \frac{\lambda^2 \sigma^2}{2} \quad [\text{Taking log}]$$

Thus,

$$\mathbb{E}[Z] \leq \frac{\log n}{\lambda} + \frac{\lambda \sigma^2}{2}$$

choose

$$\lambda = \sqrt{\frac{2 \log n}{\sigma}}$$

$$\mathbb{E}[Z] \leq \sigma \sqrt{2 \log n}$$

b. $\mathbb{P}\left(Z \geq \sqrt{2\sigma^2 \log(n/\delta)}\right) \leq \delta$ for any $\delta \in (0, 1)$.

3b) Tail bound:

$\forall$ any $a > 0$ & $\lambda > 0$, by markov,

$$P(Z \geq a) = P(e^{\lambda Z} \geq e^{\lambda a}) \leq \frac{\mathbb{E}[e^{\lambda Z}]}{e^{\lambda a}} \leq \exp\left(\log n + \frac{\lambda^2 \sigma^2}{2} - \lambda a\right)$$

Choosing $\lambda = \frac{a}{\sigma^2}$ then,

$$P(Z \geq a) \leq \exp\left(\log n - \frac{a^2}{2\sigma^2}\right)$$

Set,

$$a = \sigma \sqrt{2 \log (n/\delta)}$$

$$\Rightarrow P\left(Z \geq \sigma \sqrt{2 \log (n/\delta)}\right) \leq \delta$$

Hint: Use Jensen's inequality to show that $\exp(\lambda \mathbb{E}[Z]) \leq \mathbb{E}[\exp(\lambda Z)]$, and then provide a naive bound on the moment-generating function of Z.

Problem 4

Let $X_1, \ldots, X_n$ be independent random variables with $\mathbb{P}(X_t \leq x) \leq x$ for each $x \in [0,1]$ and $t \in [n]$. Our goal is to prove that for any $\varepsilon > n > 0$,

$$\mathbb{P}\left(\sum_{t=1}^{n} \log(1/X_t) \geq \varepsilon\right) \leq \left(\frac{\varepsilon}{n}\right)^n \exp(n - \varepsilon).$$

Hint: For X a non-negative random variable, $\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > t)\, dt$. You may find the following steps useful:

a. Show that for any λ,

$$\mathbb{P}\left(\sum_{t=1}^{n} \log(1/X_t) \geq \varepsilon\right) \leq \exp(-\lambda\varepsilon) \prod_{t=1}^{n} \mathbb{E}[\exp(\lambda \log(1/X_t))].$$

4a) $X_t \in (0,1]$

$P(X_t \leq x) \leq x, \ \forall x \in [0,1], \forall t \in [n]$

Let,

$$S := \sum_{t=1}^{n} \log\left(\frac{1}{X_t}\right)$$

For any $\lambda > 0$,

$$\mathbb{P}(S \geq \varepsilon) = \mathbb{P}(e^{\lambda S} \geq e^{\lambda \varepsilon}) \leq e^{-\lambda \varepsilon} \mathbb{E}[e^{\lambda S}]$$

$$= e^{-\lambda \varepsilon} \mathbb{E}\left[\prod_{t=1}^{n} e^{\lambda \log(1/X_t)}\right]$$

Since X_t's are independent,

$$\mathbb{E}\left[\prod_{t=1}^{n} e^{\lambda \log(1/X_t)}\right] = \prod_{t=1}^{n} \mathbb{E}\left[e^{\lambda \log(1/X_t)}\right]$$

So,

$$P(S \geq \varepsilon) \leq e^{-\lambda \varepsilon} \prod_{t=1}^{n} \mathbb{E}\left[e^{\lambda \log(1/X_t)}\right] \text{ — } ⓐ$$

b. Show that for any $0 < \lambda < 1$,

$$\mathbb{E}[\exp(\lambda \log(1/X_t))] \leq \frac{1}{1-\lambda}.$$

You may want to use the Hint to prove this.

b) Now we Bound $\mathbb{E}\left[e^{\lambda \log(1/X_t)}\right]$ $\forall\ 0<\lambda<1$

$$e^{\lambda \log(1/X_t)} = \left(1/X_t\right)^{\lambda} = X_t^{-\lambda}$$

Let,

$$Y := X_t^{-\lambda} \geq 0$$

$$\mathbb{E}[X_t^{-\lambda}] = \int_0^\infty \mathbb{P}(X_t^{-\lambda} > u)\,du = \int_0^\infty \mathbb{P}\left(X_t < u^{-1/\lambda}\right) du \quad \left[\mathbb{E}[Y] = \int_0^\infty P(Y>u)\,du\right]$$

if $0<u\leq 1$, $u^{-1/\lambda} \geq 1$ $\left[X_t \leq 1,\ \mathbb{P}\left(X_t < u^{-1/\lambda}\right) = 1\right]$

$$\therefore \int_0^1 P(X_t < u^{-1/\lambda})\,du \leq \int_0^1 1\,du = 1$$

if $u \geq 1$,

$u^{-1/\lambda} \in (0,1]$ [by assumption $P(X_t \leq x) \leq x$]

$\Rightarrow P(X_t < u^{-1/\lambda}) \leq u^{-1/\lambda}$

Thus,

$$\int_1^\infty \mathbb{P}\left(X_t < u^{-1/\lambda}\right) du \leq \int_1^\infty u^{-1/\lambda}\,du$$

for $0<\lambda<1$, this integral converges and equals

$$\int_1^\infty u^{-1/\lambda}\,du = \frac{\lambda}{1-\lambda}$$

$$\mathbb{E}\left[X_t^{-\lambda}\right] \leq 1 + \frac{\lambda}{1-\lambda} = \frac{1}{1-\lambda}.$$

$$\therefore \mathbb{E}\left[e^{\lambda \log(1/X_t)}\right] = \mathbb{E}\left[X_t^{-\lambda}\right] \leq \frac{1}{1-\lambda} \quad \forall\ 0<\lambda<1. \rightarrow \text{b}$$

c. Choose an appropriate λ to complete the proof.

c) Choosing λ

$$P(S \geq \varepsilon) \leq e^{-\lambda\varepsilon}\left(\frac{1}{1-\lambda}\right)^n, \quad 0<\lambda<1 \quad [\text{b} \rightarrow \text{a}]$$

Now assuming $\varepsilon > n$, we choose

$$\lambda = 1 - \frac{n}{\varepsilon} \in (0,1)$$

Then,

$$1-\lambda = \frac{n}{\varepsilon},$$

$$\left(\frac{1}{1-\lambda}\right)^n = \left(\frac{\varepsilon}{n}\right)^n,$$

$$e^{-\lambda\varepsilon} = e^{-(1-n/\varepsilon)\varepsilon} = e^{-(\varepsilon-n)} = e^{n-\varepsilon}.$$

$$\therefore \mathbb{P}\left(\sum_{t=1}^n \log\frac{1}{X_t} \geq \varepsilon\right) \leq \left(\frac{\varepsilon}{n}\right)^n e^{n-\varepsilon}, \quad \varepsilon > n$$

Problem 5 (Bonus)

Suppose that X is σ-subgaussian. Prove that $\mathbb{E}[X] = 0$ and $\mathrm{Var}[X] \leq \sigma^2$.

Your solution here (add `output/5.jpeg` when ready).